

M R KHAN DIPU

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LinkedIn | GitHub | Portfolio

PROFESSIONAL SUMMARY

Software Engineering graduate with completed BSc and MSc degrees in Data Science, with experience across research, administration, and web development. Interested in research opportunities to continue developing expertise in AI and machine learning, administrative and executive roles to strengthen interpersonal and professional skills, and web development to build applications that integrate machine learning models. Open to research, administrative, executive, and developer opportunities.

WORK EXPERIENCE

Executive, Academic Management | The DrAcademy 08/2025 - 06/2026

Medical coaching center for UK Royal College courses, PLAB, and OET.

- Focused primarily on digital marketing: wrote SEO-friendly advertising copy for Google and Facebook campaigns, YouTube content, and website blog posts.
- Applied basic SEO practices, including meta descriptions, heading structuring, avoiding repeated H1 headings, image alt text, and content ideation.
- Managed student relationships and mentor communication.
- Oversaw eLearning management, class management, and study materials preparation.
- Supported HR and administration functions.

Role concluded due to company budget cuts.

Batch Coordinator, Q&A | Udvash 10/2023 - 01/2024

- Monitored teacher interaction with students and assured quality of teacher responses.
- Administered teachers and prepared payment schemes for teaching staff.
- Conducted teacher training and teacher recruitment.

RESEARCH EXPERIENCE

ML/AI Researcher (Independent, Unpaid, Remote) | Group of Bio-Photomatix 02/2024 - 07/2026

Independent research work conducted remotely alongside professional employment.

- Performed brain signal processing, image data processing, and feature extraction for biomedical research.
- Carried out medical data modeling and parameter tuning for machine learning models.
- Contributed to manuscript writing for research publications.

Research Areas: Biomedical Signal Processing, EEG Signal Processing, Machine Learning, Deep Learning, Medical Data Modeling, Healthcare AI

PUBLICATIONS & CONFERENCES

Lightweight 1D Convolutional Neural Network Architecture for Detecting Epileptic Seizures

4th International Conference on Intelligent Cyber Physical Systems and Internet of Things (ICoICI 2026), 1-3 September 2026.

Gastrointestinal Health: A Convolutional Neural Network Approach for Gastrointestinal Disorders Prediction for Disease Diagnosis

16th International IEEE Conference on Computing, Communication and Networking Technologies (ICCCNT), 2025.

PROJECTS

Bazar Dor

Bengali market-price web application that allows users to explore product prices and browse products by category. Users can view current and historical price data, track price changes, and access a responsive interface with user authentication.

Technologies: Next.js, React, Tailwind CSS, MongoDB, better-auth

Live Site | GitHub

Bangla News 24

Bengali news portal featuring live headlines, category-based news browsing, a Most Read section, and a scrolling news ticker. Includes user authentication with email/password and Google sign-in, email verification, password reset, and protected routes.

Technologies: Next.js, TypeScript, Tailwind CSS, better-auth, MongoDB, Resend

Live Site | GitHub

FitLog

Responsive gym-companion web application for browsing a workout library and building a daily workout plan. Users can add workouts to a plan or save them for later, track exercises, minutes and calories, and sort saved items; data persists in the browser via local storage.

Technologies: Next.js, React, Tailwind CSS, DaisyUI, React Context API, React Toastify, React Icons

Live Site | GitHub

Alzheimer Dementia Classifier

Deep learning-powered MRI classifier for dementia severity classification.

Technologies: Streamlit, TensorFlow, Keras

Live Site | GitHub

Quantum Circuit Explorer

Interactive web app fro building, simulating, and visualizing circuits, including circuit diagrams, measurement histograms, and Bloch sphere representations.

Technologies: Streamlit, Qiskit, Qiskit Aer, Matplotlib

Live Site | GitHub

TECHNICAL SKILLS

Programming: Python

Web Development: HTML, CSS, Tailwind CSS, JavaScript, TypeScript, React, Next.js, Node.js, Express.js

Data Science & Machine Learning: Python, Pandas, NumPy, Scikit-learn, Machine Learning, Deep Learning, Natural Language Processing, EEG Signal Processing

Tools: Git, GitHub, Canva, Microsoft Office

Soft Skills: Leadership, Communication, Teamwork, Problem-solving, Administration and Management

EDUCATION

MSc in Software Engineering, Major in Data Science 2023 - 2025

Daffodil International University, Dhaka, Bangladesh | CGPA: 3.68/4.00

Thesis: "Major Depressive Disorder Diagnostics: Early Detection Through Advanced Deep Learning Analysis of Physiological Signal" (Thesis GPA: 4.00)

BSc in Software Engineering, Major in Data Science 2019 - 2023

Daffodil International University, Dhaka, Bangladesh | CGPA: 3.65/4.00

Thesis: "Research on Psychological Impact for Excessive Using of Smart Device with Respect to Social Media Addiction Using Artificial Neural Network" (Thesis GPA: 3.75)

CERTIFICATIONS

Analytics Decision Making, HRDI, DIU — Data Analysis, Visualization, Modeling

Responsive Web Design, freeCodeCamp — HTML, JavaScript, Bootstrap

Office Application Course, IT Solution — Microsoft Word, Excel, PowerPoint, Outlook, Access

RELEVANT COURSEWORK

Machine Learning Driven Data Analysis, Statistical Data Analysis, Data Science Major Capstone Project, Applied Machine Learning, Research Design and Applications for Data and Analysis, Data Visualization and Communication, Artificial Intelligence, Object Oriented Design and Concepts, Software Project Management

EXTRACURRICULAR ACTIVITIES

- Participated in the 3MT Thesis Competition at the National Data Science & AI Startup Bootcamp 2024, DIU (2024).
- Attended the Workshop on Data Science and Applied Machine Learning at DIU (2024).
- Member of the Data Science Club, Daffodil International University.
- Represented DIU at Teknofest-2022 Online through International Affairs (2022).
- Worked with Daffodil Prothom Alo Bondhushova and performed as a "Nut Seller" in Choruivati (2021).
- Volunteered with the DIU Voluntary Service Club, including admission-test support (2021).
- Assisted in decorating and performing at the Freshers Reception Program (2019).
- Worked with the Daffodil Photography Club, exploring creative photography projects.

AWARDS & ACHIEVEMENTS

- Merit Award (2013) for meritorious result in the Junior School Certificate Examination 2013.

REFERENCES

Imran Mahmud

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